

Predicting Diabetes: A Machine Learning Approach

Introduction

Diabetes is a chronic condition affecting millions globally, resulting from the body's inability to produce or use insulin effectively. This leads to elevated blood sugar levels, which can cause significant health complications if not managed early. This project aims to predict the likelihood of diabetes using various machine learning algorithms. By applying logistic regression, decision trees, random forest classifiers, and Naive Bayes algorithms, we aim to determine whether a person has diabetes accurately.

Dataset

Source

The dataset used for this project is sourced from Kaggle and was originally provided by the National Institute of Diabetes and Digestive and Kidney Diseases. We acknowledge and appreciate their contribution in making this dataset publicly available for research purposes.

[<https://www.kaggle.com/datasets/mathchi/diabetes-data-set>]

Data Overview

The dataset comprises various features relevant to diabetes prediction. It includes attributes such as age, glucose levels, blood pressure, and BMI.

Data Collection and Preprocessing

Data Collection

The dataset was collected in its cleaned format, containing relevant features necessary for the prediction task.

Data Preprocessing

Data preprocessing involved several steps:

- **Handling Missing Values:** Missing values were imputed using appropriate statistical methods.
- **Feature Scaling:** Features were scaled to ensure uniformity in the data.

- **Encoding Categorical Variables:** Categorical variables were encoded into numerical formats.

Data Analysis and Exploration

Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset's structure and relationships between features. Key insights included:

- **Distribution of Diabetes:** Analysis of the proportion of diabetic and non-diabetic instances.
- **Correlation Analysis:** Identification of significant correlations between features and the target variable.

Model Development

Model Selection

We evaluated four machine learning algorithms:

- **Logistic Regression:** A statistical model used for binary classification.
- **Decision Tree Classifier:** A tree-like model used for making decisions based on feature values.
- **Random Forest Classifier:** An ensemble method that combines multiple decision trees to improve prediction accuracy.
- **Naive Bayes Classifier:** A probabilistic classifier based on Bayes' theorem.

Model Training and Evaluation

Models were trained using a training dataset and evaluated on a separate test set. Performance metrics included accuracy, precision, recall, and F1-score.

Model Tuning

Hyperparameters for each model were tuned to enhance performance:

- **Logistic Regression:** Adjusted regularization parameters.
- **Decision Tree:** Optimized tree depth and split criteria.
- **Random Forest:** Tuned the number of trees and maximum features.
- **Naive Bayes:** Adjusted smoothing parameters.

Model Interpretation

Important Factors

Key factors influencing diabetes prediction were identified through feature importance analysis. Notable features included glucose levels, BMI, and age.

Feature Selection

Features were selected based on their relevance and impact on the model's performance. Irrelevant or redundant features were removed to improve the model's efficiency.

Conclusion

This project successfully demonstrated the application of various machine learning algorithms in predicting diabetes. By leveraging logistic regression, decision trees, random forest classifiers, and Naive Bayes, we achieved reliable predictions and identified critical features affecting diabetes risk. Future work may involve exploring additional algorithms or integrating external datasets to enhance prediction accuracy further.